

Partial solution to the exam (10.2.00)

1 Multiple Choice Questions

1. If $G'(x) = \frac{1}{x(x^2+1)}$ such that $G(1)=1$, then $G(2) = \ln(2^{3/2}(\sqrt{5})^{-1}) + 1$.
2. Let $f(x) = \frac{\ln(1+x)}{x^{3/2}}$. Note that near $x = 0$, $\ln(1+x)$ behaves like x . Thus, $f(x)$ behaves like $x^{-1/2}$ near $x = 0$. Hence, $\lim_{x \rightarrow 0^+} f(x) = \infty$ and the integral $\int_0^1 f(x)dx$ converges. Also note that for x large enough, $\ln(1+x) \leq x^{1/4}$. Thus, $\int_1^\infty f(x)dx$ converges.
Next, $f(x)$ is positive in $(0, \infty)$, but $\lim_{x \rightarrow \infty} f(x) = 0$. Therefore, the minimum of f is not attained in $[1, \infty)$.
Finally, for n large enough, the sequence $(nf(n))$ is non increasing and tends to 0. Thus, $\sum_1^\infty (-1)^n nf(n)$ converges.
3. The tangent equation is $y = -x + 1$.
4. Let $f(x) = ax + bx^2 + R_2(x)$. By the definition of Taylor's polynomial, $f(0) = 0$, $f'(0) = a$, $f''(0) = 2b$. It is easy to see that if, for example, $f^{(3)}(x)$ is continuous then

$$\lim_{x \rightarrow 0} \frac{R_2(x)}{x^3} = \lim_{x \rightarrow 0} \frac{f^{(3)}(c(x))}{3!} = \frac{f^{(3)}(0)}{3!},$$

therefore, the limit may exist.

It is easy to see that $\lim_{x \rightarrow 0} \frac{\ln(1+f(x))}{x^2} = b$, which implies that $\lim_{x \rightarrow 0} (1+f(x))^{1/x^2} = e^b$, which equals to 1 if and only if $b = 0$.

Note that $\lim_{x \rightarrow 0} \frac{f(x)}{x} = a$. Hence, if $a = 1$ then $\lim_{n \rightarrow \infty} \frac{f(\frac{1}{n})}{\frac{1}{n}} = 1$. Thus, $\sum_1^\infty f(\frac{1}{n})$ diverges.

Finally, to see the $g(x)$ is differentiable,

$$\lim_{x \rightarrow 0} \frac{\frac{f(x)}{x} - 1}{x} = \lim_{x \rightarrow 0} \frac{f(x) - x}{x^2} = \lim_{x \rightarrow 0} \frac{bx^2 + R_2(x)}{x^2} = b$$

5. Let $f(x) = x^4 - 2x^3 + 2x^2 + cx + 1$ and set $g(x) = 1/f(x)$ for points in which $f(x) \neq 0$, and 1 otherwise. Since f is a nontrivial polynomial then it is not bounded in $(-\infty, 0]$. Also, since $f(0) = -1$ and f tends to infinity as $x \rightarrow \infty$, then there is some $x_0 \in (0, \infty)$ such that $f(x_0) = 0$. Hence, $g(x)$ can not be bounded there.

If $c = 0$ then $f'(x) = 4x^3 - 6x^2 + 4x$. Thus, if $|x| \leq 1$ then $|f'(x)| \leq 14$. Therefore, since $f(0) = -1$ f can not have zeros in $(-1/14, 1/14)$.

Since $f''(x) = 12x^2 - 12x + 4 \neq 0$ for every x , then by Rolle's Theorem f' can have at most a single zero, implying that f can have two zeros at most.

Finally, we saw that there is some $x_0 \in [0, \infty)$ such that $f(x_0) = 0$. Therefore, $f(x) = (x - x_0)^d q(x)$, where $d \geq 1$ is an integer and $q(x)$ is a polynomial which has no zeros in a neighborhood of x_0 . Since q is continuous it maintains a constant sign in a neighbourhood of x_0 , and we may assume that it is strictly positive. Therefore,

$$\lim_{x \rightarrow x_0} \frac{g(x)}{(x - x_0)^d} = q(x_0) > 0.$$

Since $\int_{x_0-\delta}^{x_0+\delta} \frac{1}{(x-x_0)^d}$ diverges, so does $\int_{x_0-\delta}^{x_0+\delta} g(x)$.

2 Yes-No Questions

Here is the solution to some of the Yes-No Questions.

1. If f is monotone in $[a, b]$ then it attains its minimum and maximum in a and b . Also, it was shown that if f is monotone in $[a, b]$ then it is Riemann integrable there.
2. If $\sum a_n$ converges, then (a_n) tends to 0. Thus, every subsequence (a_{n_j}) also tends to 0. On the other hand, $\sum a_{n_j}$ may diverge. For example, let $a_n = (-1)^n/n$ and put $n_j = 2j$. In this case, $\sum a_{n_j} = \sum 1/2n = \infty$.
3. Let $a_{2n} = 1/n$ and $a_{2n+1} = 1/(2n+1)^2$. Then (a_n) tends to 0 but a_{n+1}/a_n does not converge at all.
4. If $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $|f'(x)| < 1$, and (a_n) is increasing such that $f(a_n) = 1 + (-1)^n/n$, then by Lagrange's Theorem,

$$\left| \frac{f(a_{n+1}) - f(a_n)}{a_{n+1} - a_n} \right| \leq 1.$$

On the other hand, $|f(a_{n+1}) - f(a_n)| \geq \frac{2}{n+1}$.

Thus, $a_{n+1} - a_n = |a_{n+1} - a_n| \geq \frac{2}{n+1}$. But,

$$a_{n+1} = a_1 + (a_2 - a_1) + \dots + (a_{n+1} - a_n) \geq a_1 + \sum_{i=1}^n \frac{2}{i+1},$$

and since $\sum 1/n = \infty$ then (a_n) is not bounded, which implies that it must converge to infinity.

5. The convergence radius of $\sum (\frac{2^n}{n} + \frac{3^n}{n^2})x^n$ is $1/3$.

3 Open Questions

Below are partial solutions to the open part + the grading instructions. In some cases we allowed ourselves to slightly alter the guidelines and to add or subtract up to two points based on the general impression of the answer.

1. If $a_n = n^n$ then $\sum a_n x^n$ converges only for $x = 0$ (there are many other examples). We gave 5 points for a correct example (even without an explanation) and no points if the example was wrong.
2. If such a function f where to exist, then by Lagrange's Theorem there would be some $c \in (0, 5)$ such that

$$2 = \frac{f(5) - f(0)}{5 - 0} = f'(c) < 2,$$

which is impossible.

For noting that such a function does not exist we gave 1 point. For a complete proof we gave 5 points. WE DIDN'T AWARD MORE THAN 1 POINT FOR AN "INTUITIVE" EXPLANATION!!!.

3. If $f(x) = (1 - x)^{-1/2}$ for $x \neq 1$ and 0 for $x = 1$, then f is not Riemann integrable in $[0, 1]$ because it is not bounded there. However $\int_0^1 f(x) dx$ converges. We gave 5 points for a correct example and none if the example was wrong.

A common error was to select $f(x) = (1 - \sqrt{x})^{-1}$. This is wrong since near $x = 1$ this function behaves like $2/(1 - x)$, which has a diverging integral!

4. $\sum n^{-(1+1/n)}$ diverges by the comparing it to the diverging $\sum \frac{1}{n}$. We gave 2 points for identifying that the correct rate by which a_n tended to 0 was $1/n$, and additional 3 points for doing the arithmetic. The final 2 points were awarded for knowing that $\sum 1/n$ diverges. If the score was ≥ 4 the student was considered successful on the homework question.
5. f_n converges uniformly to f in $[0, 1]$ if for every $\varepsilon > 0$ there is some integer N , such that for every $n > N$ and every $x \in [0, 1]$, $|f_n(x) - f(x)| < \varepsilon$.
another option is this: for every $\varepsilon > 0$ there is some integer N such that for every $n > N$

$$\sup_{x \in [0, 1]} |f_n(x) - f(x)| < \varepsilon.$$

The grading: A mistake with even a single quantifier results in a score of 0/7!. If one forgot the "for every $x \in [0, 1]$ ", or did not say what the supremum was taken with respect to, the score was 2/7.

6. The proof: Since $a_n \rightarrow \infty$, then for every M there is some integer N such that for every $n > N$, $a_n > M$. We will construct the subsequence inductively: let $n_1 = 1$. By the above definition of the limit, there is some integer N_2 such that for every $n > N_2$, $a_n > a_1$. Select $n_2 = N_2 + 1$.

Turning to the induction step, suppose we have constructed $a_1 < \dots < a_{n_j}$ such that $n_1 < n_2 < \dots < n_j$. Since $a_n \rightarrow \infty$, there is some index N_{j+1} such that for every $n > N_{j+1}$, $a_n > a_{n_j}$. Let $n_{j+1} = \max\{N_{j+1} + 1, n_j + 1\}$. Thus, $n_{j+1} > n_j$ and $a_{n_{j+1}} > a_{n_j}$.

The grading: For using the fact that if $a_n \rightarrow \infty$ then for every M , $a_n > M$ if n is large enough - 4 points. For constructing a_{n_j} such that $a_{n_{j+1}} \geq a_{n_j}$, additional 6 points. The final 4 points are awarded if one proves that $n_j > n_{j+1}$ - which WAS NOT done by most of the students.